

NOVOBIRILYUSSY, Russian Federation

WMO: 293670

Lat: 56.9697N

Lon: 90.6831E

Elev: 181

StdP: 99.17

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.9	(c) -35.2	(d) -42.2	(e) 0.1	(f) -39.1	(g) -37.8	(h) 0.1	(i) -34.8	(j) 6.1	(k) -7.1	(l) 5.4	(m) -7.1	(n) 0.5	(o) 190	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 29.0	(d) 19.4	(e) 27.1	(f) 18.6	(g) 25.3	(h) 17.8	(i) 20.9	(j) 26.5	(k) 19.9	(l) 25.1	(m) 19.0	(n) 23.8	(o) 2.1	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.1	(b) 14.2	(c) 23.3	(d) 18.1	(e) 13.3	(f) 22.5	(g) 17.1	(h) 12.4	(i) 21.5	(j) 61.7	(k) 26.5	(l) 58.1	(m) 25.3	(n) 54.9	(o) 23.9	(p) 24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.0	(b) 5.0	(c) 4.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 6.0	(b) 5.0	(c) 4.3	(e) -43.1	(f) 22.4	(g) 3.4	(h) 1.6	(i) -45.5	(j) 23.6	(k) -47.5	(l) 24.5	(m) -49.4	(n) 25.4	(o) -51.9	(p) 26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-17.3	-13.3	-6.6	2.6	9.0	15.9	18.2	15.2	8.0	0.4	-10.1	-14.5		
	DBStd	13.89	9.59	7.95	6.97	4.71	5.47	4.31	3.17	3.72	4.52	5.45	9.18	9.33		
	HDD10.0	4093	846	652	513	226	88	7	0	4	92	300	604	761		
	HDD18.3	6512	1104	886	772	471	294	93	41	109	312	557	854	1019		
	CDD10.0	698	0	0	0	4	56	186	255	166	30	2	0	0		
	CDD18.3	75	0	0	0	0	3	22	38	12	0	0	0	0		
	CDH23.3	830	0	0	0	2	50	316	350	102	9	0	0	0		
	CDH26.7	176	0	0	0	0	8	75	73	20	0	0	0	0		
(14) Wind	WSAvg	1.9	2.0	2.0	2.1	2.4	2.2	1.6	1.4	1.4	1.7	2.1	2.0	2.0		
(15) Precipitation	PrecAvg	510	25	18	18	27	43	63	56	76	49	50	42	38		
	PrecMax	648	52	33	47	43	90	118	118	155	102	80	65	53		
	PrecMin	415	3	8	6	10	15	19	19	22	9	22	23	14		
	PrecStd	66	11	6	10	11	20	26	26	32	23	14	10	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	1.6	9.0	20.6	28.0	30.8	31.1	29.3	24.8	16.6	8.5	1.4		
		MCWB	-1.1	-1.0	4.0	11.0	16.0	20.1	20.9	20.2	15.8	9.4	6.0	-0.1		
	2%	DB	-1.2	0.3	6.0	16.3	24.6	28.7	29.2	26.5	20.7	12.4	5.5	-0.2		
		MCWB	-2.6	-1.7	1.8	8.5	14.4	18.8	20.0	19.3	14.8	7.7	3.7	-1.2		
	5%	DB	-2.9	-1.1	4.4	13.3	21.9	26.8	27.2	24.1	17.9	10.0	3.4	-1.8		
		MCWB	-4.0	-2.5	1.1	7.1	12.7	17.8	19.8	18.1	12.9	6.7	1.4	-2.9		
	10%	DB	-4.6	-2.4	2.7	10.8	19.1	24.7	25.5	22.3	15.2	7.9	1.4	-3.5		
		MCWB	-5.6	-3.9	0.1	5.7	11.6	17.0	18.7	17.2	11.5	5.4	-0.6	-4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	-0.2	4.3	11.5	16.8	21.6	22.7	21.5	16.6	10.1	6.3	0.2		
		MCDB	0.6	1.2	7.2	19.9	26.1	27.8	27.6	28.0	21.6	14.3	8.0	1.1		
	2%	WB	-2.5	-1.4	2.5	9.3	15.2	20.1	21.5	19.9	15.1	8.5	3.8	-1.1		
		MCDB	-1.5	-0.2	5.3	15.2	21.7	26.5	26.7	24.5	20.4	11.7	5.2	-0.4		
	5%	WB	-4.2	-2.5	1.4	7.6	13.8	19.0	20.5	18.8	13.7	7.1	1.5	-2.9		
		MCDB	-2.9	-1.1	3.8	12.7	20.7	24.8	25.5	23.0	17.6	9.5	3.0	-2.0		
	10%	WB	-5.6	-3.9	0.2	5.8	12.4	17.7	19.5	17.7	12.2	5.8	-0.1	-4.5		
		MCDB	-4.6	-2.5	2.6	10.2	18.3	23.2	24.2	21.4	14.7	8.1	1.0	-3.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.2	10.5	9.8	10.8	11.6	10.9	9.9	8.9	6.7	6.9	7.4		
		MCDBR	7.3	7.4	10.9	13.8	15.2	16.0	13.8	12.8	14.3	9.9	6.8	6.8		
		MCWBR	6.5	6.3	7.8	7.8	7.1	7.1	6.5	6.8	7.9	6.2	5.5	6.3		
		MCDBR	7.2	6.4	9.1	13.0	13.5	13.9	11.5	11.3	13.1	8.7	6.4	6.0		
	5% WB	MCWBR	6.7	5.7	6.8	7.8	6.8	6.5	6.1	6.4	7.5	5.9	5.3	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.322	0.366	0.395	0.408	0.409	0.432	0.410	0.370	0.349	0.316	0.287		
	taud		1.979	1.966	1.916	2.057	2.206	2.291	2.232	2.271	2.334	2.221	2.016	1.930		
	Ebn at Noon		534	683	755	809	827	832	801	792	770	664	490	408		
	Edh at Noon		65	101	138	141	130	121	126	114	92	78	62	52		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.65	2.96	4.32	4.96	5.92	5.40	4.20	2.64	1.32	0.65	0.40		
	RadStd		0.09	0.15	0.22	0.31	0.55	0.54	0.36	0.36	0.35	0.13	0.11	0.07		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page